UNIT 6 · TOPIC 8

Evaluation and Prediction of Traction Performance

Syllabus topic: Evaluation and prediction of traction performance

The cone-index wheel numeric (Freitag; Wismer & Luth 1974)

$$\text{Wheel numeric } Cn = C \cdot b \cdot d \,/\, W$$

C = cone index (soil strength); b = tyre section width; d = tyre overall diameter; W = dynamic wheel load. A higher Cn means stronger soil or a lighter load — better traction.

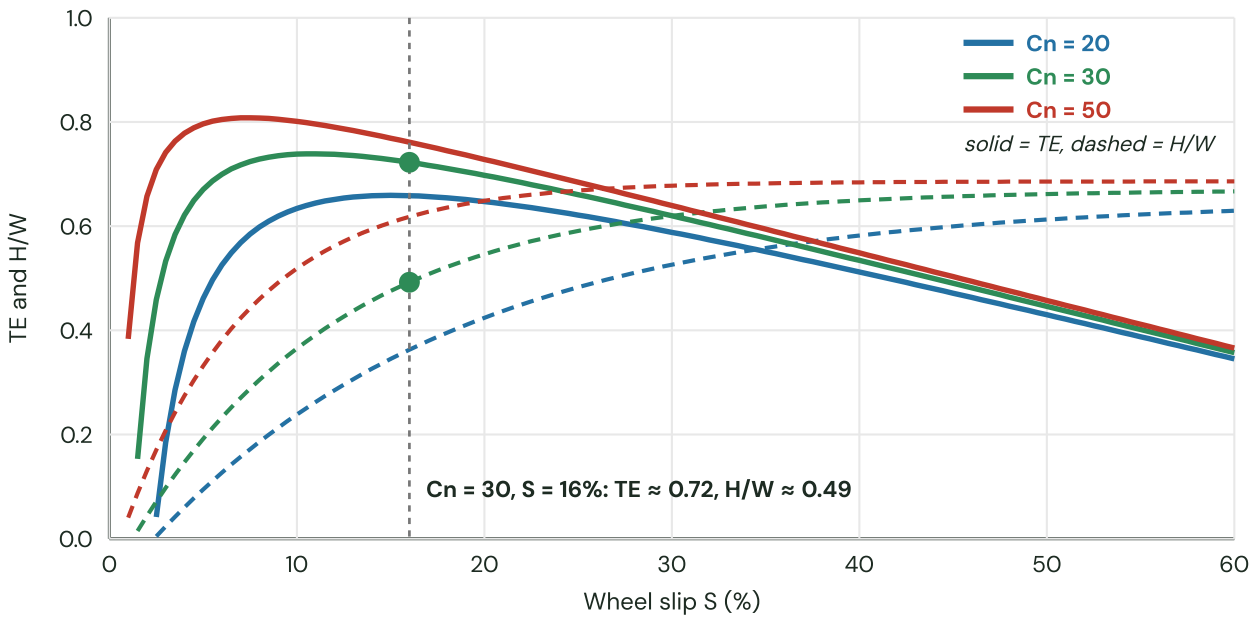
$$\text{Gross traction coefficient: } \mu_g = 0.75(1 - e^{(-0.3 \cdot Cn \cdot S)})$$

$$\text{Motion-resistance ratio: } \rho = 1.2/Cn + 0.04$$

$$\text{Net pull/weight: } H/W = 0.75(1 - e^{(-0.3 \cdot Cn \cdot S)}) - (1.2/Cn + 0.04)$$

Corrected term. The source notes print the motion-resistance term as (1/Cn + 0.04). The published Wismer–Luth (1974) relation is ρ = 1.2/Cn + 0.04, and only that form reproduces the textbook design example below (TE = 0.72). Exam options may show either; the structure — gross traction minus motion resistance — is what is tested.

Tractive efficiency vs slip (p. 250)



TE (solid) and H/W (dashed) vs slip from the Wismer–Luth equations (redrawn with the 1.2/Cn + 0.04 term)

$$TE = (\mu/\mu_g) \cdot (1 - S)$$

- TE rises to a MAXIMUM at low slip, then falls. Maximum TE occurs at LOWER slip for large Cn (strong soil / light load).
- Maximum wheel POWER output occurs at the slip of maximum TE, but H/W there is not near its maximum — so for large drawbar pull the DESIGN SLIP is set to the RIGHT of the TE peak.

Design example (p. 250) — memorise: Cn = 30 → TE = **0.72**, H/W = **0.51**, slip = **0.16**.
Check from the equations: μ_g = 0.75(1 - e^(-0.3 × 30 × 0.16)) = 0.75(1 - e^(-1.44)) = **0.572**; ρ = 1.2/30 + 0.04 = **0.08**; H/W = 0.572 - 0.08 = **0.49**; TE = (0.49/0.572)(1 - 0.16) = **0.72**. (The textbook's 0.51 is read from its design chart.)

Firm-surface grounding example (p. 250). Burt et al. (1979) validated the relations on a **12.4-28 bias R-1** tyre in Decatur clay loam at a mean cone index of **252.5** over 0–15 cm. (The source prints the unit as "N/cm"; cone index is a pressure, so N/cm² is presumably meant.) Compacted dry clay gives a large Cn.

Evaluating traction performance

1. Test on a traction TEST CAR / single-wheel tester, or in the field.
2. Record dynamic load W, drawbar pull H, axle torque T, travel speed and wheel speed (→ slip).
3. Plot pull–slip, torque–slip, TE–slip and H/W curves for each tyre/soil/load combination.

- OECD/Nebraska drawbar tests report maximum pull at a stated slip limit (**15%** — Unit 4, Topic 11).
- Predicted performance from Cn lets designers BALANCE the vehicle across the range of soil strengths it will meet.

Practice MCQs

Q89. The dimensionless wheel numeric (cone-index number) is:
(a) Cn = C·b·d/W (b) Cn = W/Cbd
(c) Cn = C/W (d) Cn = bd/C

Q90. In Cn = Cbd/W, C is the:
(a) Cohesion (b) Cone index (soil strength)
(c) Circumference (d) Camber

Q91. A higher Cn value indicates:
(a) Weaker soil (b) Stronger soil or lighter load → better traction
(c) More slip (d) More sinkage

Q92. The Wismer–Luth net-pull relation is H/W =
(a) 0.75(1 - e^(-0.3CnS)) - (1/Cn + 0.04) (b) CnS
(c) 0.75CnS (d) 1 - S

Q93. The term (1/Cn + 0.04) in that equation represents the:
(a) Gross traction (b) Motion-resistance ratio
(c) Slip (d) Sinkage

Q94. The gross traction coefficient in the Wismer–Luth model is:
(a) 0.75(1 - e^(-0.3CnS)) (b) 1/Cn
(c) 0.04 (d) CnS

Q95. Tractive efficiency with slip:
(a) Rises indefinitely (b) Peaks at low slip then falls
(c) Is constant (d) Is zero at low slip

Q96. Maximum TE occurs at LOWER slip for:

(a) Small Cn
(b) Large Cn (strong soil/light load)
(c) Wet clay
(d) High inflation

Q97. For large drawbar pull, the design slip is chosen:

(a) At the TE peak
(b) To the right of the TE peak
(c) At zero slip
(d) At 100% slip

Q98. In the standard design example, Cn = 30 gives approximately:

(a) TE 0.72, H/W 0.51, slip 0.16
(b) TE 0.30
(c) TE 0.95
(d) TE 0.10

Answers

Q89 — (a) Cn = C·b·d/W — *pp. 248–250 — Cn = Cbd/W.*

Q90 — (b) Cone index (soil strength)

Q91 — (b) Stronger soil or lighter load → better traction

Q92 — (a) 0.75(1 - e^(-0.3CnS)) - (1/Cn + 0.04) — *Option (a) is the intended answer. The published Wismer–Luth motion-resistance term is 1.2/Cn + 0.04 (see the note in Topic 8).*

Q93 — (b) Motion-resistance ratio — *ρ ≈ 1.2/Cn + 0.04 in the published form; the source prints 1/Cn + 0.04.*

Q94 — (a) 0.75(1 - e^(-0.3CnS)) — *μ_g = 0.75(1 - e^(-0.3CnS)).*

Q95 — (b) Peaks at low slip then falls — *p. 250 — TE peaks at low slip.*

Q96 — (b) Large Cn (strong soil/light load) — *Lower slip for larger Cn.*

Q97 — (b) To the right of the TE peak — *Design slip right of the peak for pull.*

Q98 — (a) TE 0.72, H/W 0.51, slip 0.16 — *Cn = 30 → TE 0.72, H/W 0.51, slip 0.16 (p. 250).*

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units, Ch. 10, pp. 240–252; Wismer & Luth (1974), Off-road traction prediction for wheeled vehicles, Trans. ASAE; Freitag (1965) mobility-number dimensional analysis; Bekker pressure–sinkage theory; ASAE S296.2 (slip/rolling radius), S313.2 (cone index); NPTEL traction lectures.